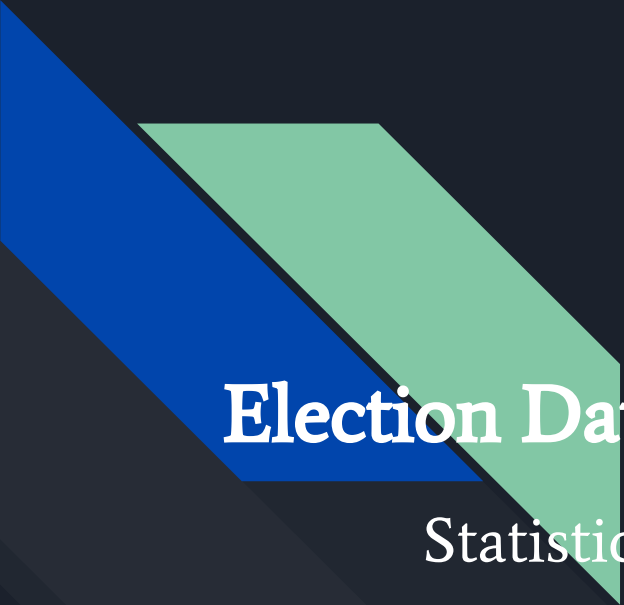




Final Project

Election Data Analysis & Visualization – 2019

Statistical Analysis, Tableau & SQL (BigQuery)

By Gayatri Thombare



Introduction

- This project analyzes the Indian General Election 2019 data.
- The aim is to understand voting patterns and party performance.
- Five states were selected: Gujarat, Karnataka, Maharashtra, Rajasthan, and Tamil Nadu.
- The analysis uses statistical charts, Tableau visualizations, and SQL (BigQuery).
- This study helps identify election trends and competitiveness across states.

Part 1:- Statistical Analysis

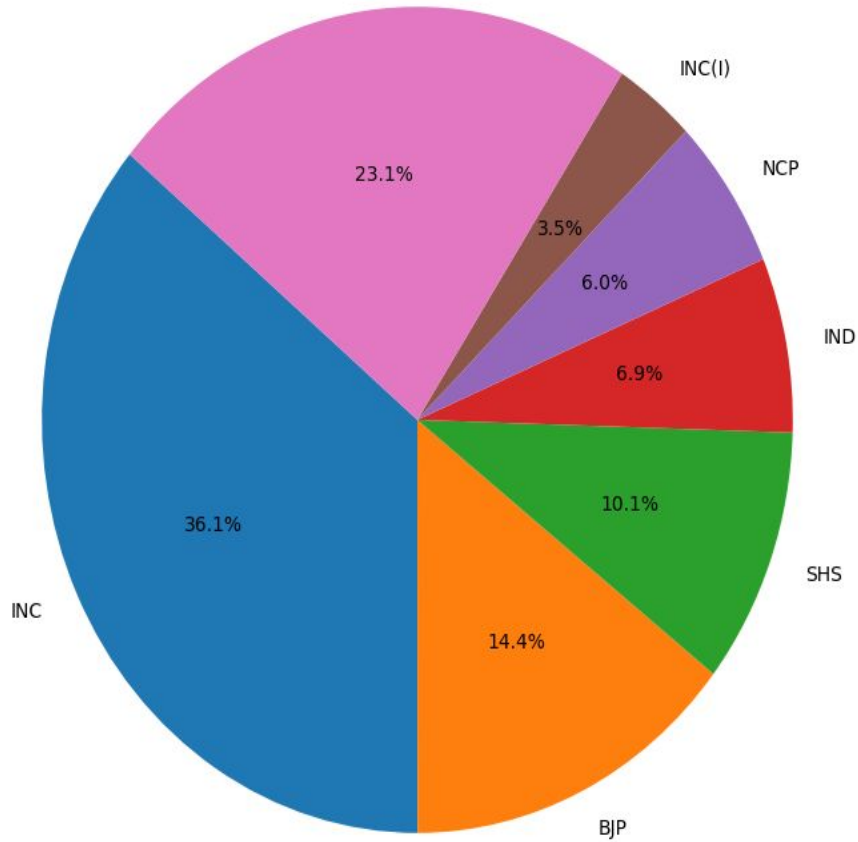


Statistical analysis is performed using pie charts to show party-wise vote percentages in selected states during the 2019 Lok Sabha elections. It helps in understanding vote distribution, party dominance, and competition levels. This analysis provides clear insights into voter behavior and electoral patterns.

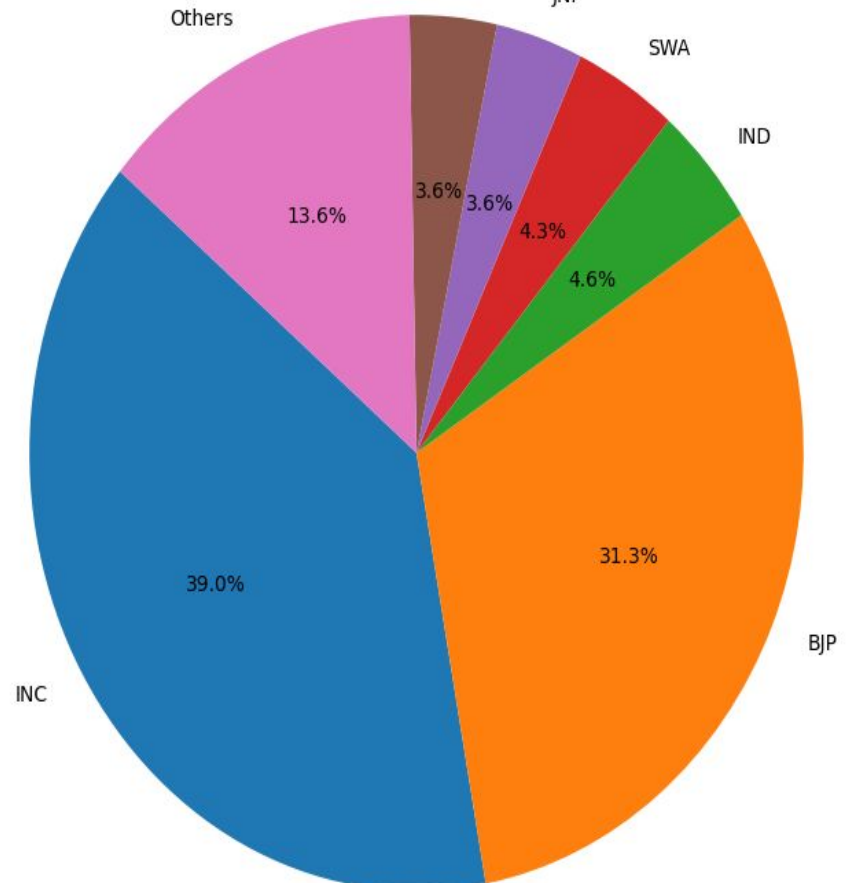
1. Party-wise Vote Percentage (Pie Charts)

Party-wise vote percentage is analysed using pie charts to represent the distribution of votes among different political parties in each selected state. These charts help in identifying the leading party and understanding the level of competition between major parties. The analysis also highlights the presence of multi-party participation and vote fragmentation in certain states. Overall, this visualization provides a clear and comparative view of party performance and voter preferences across states.

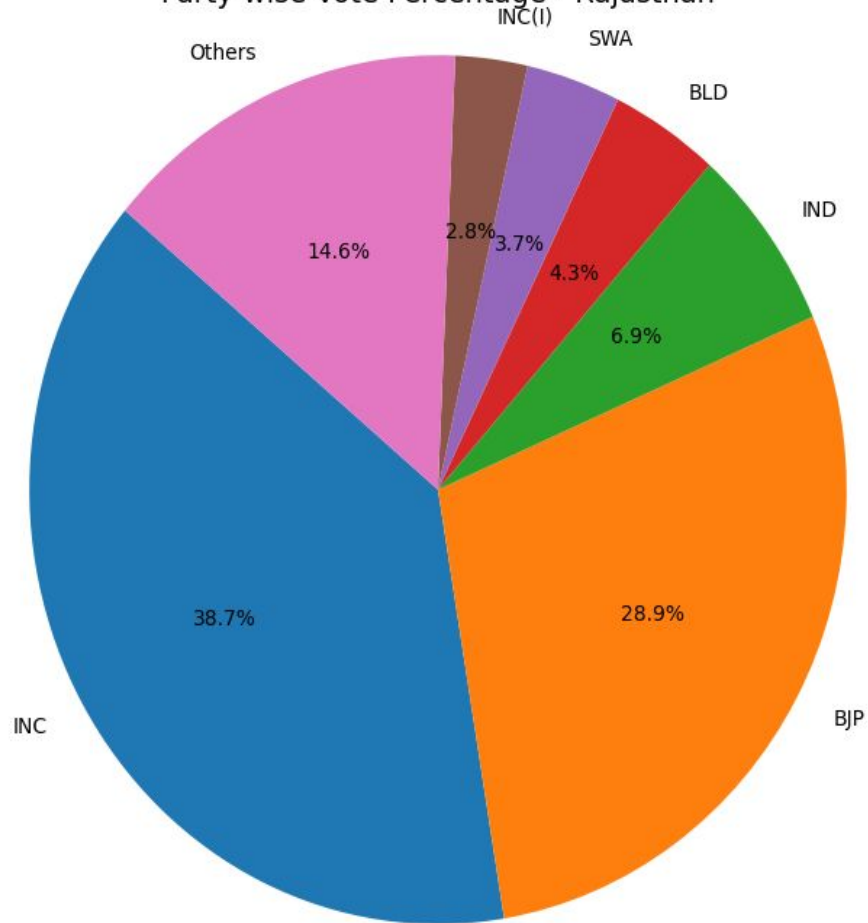
Party-wise Vote Percentage - Maharashtra
Others



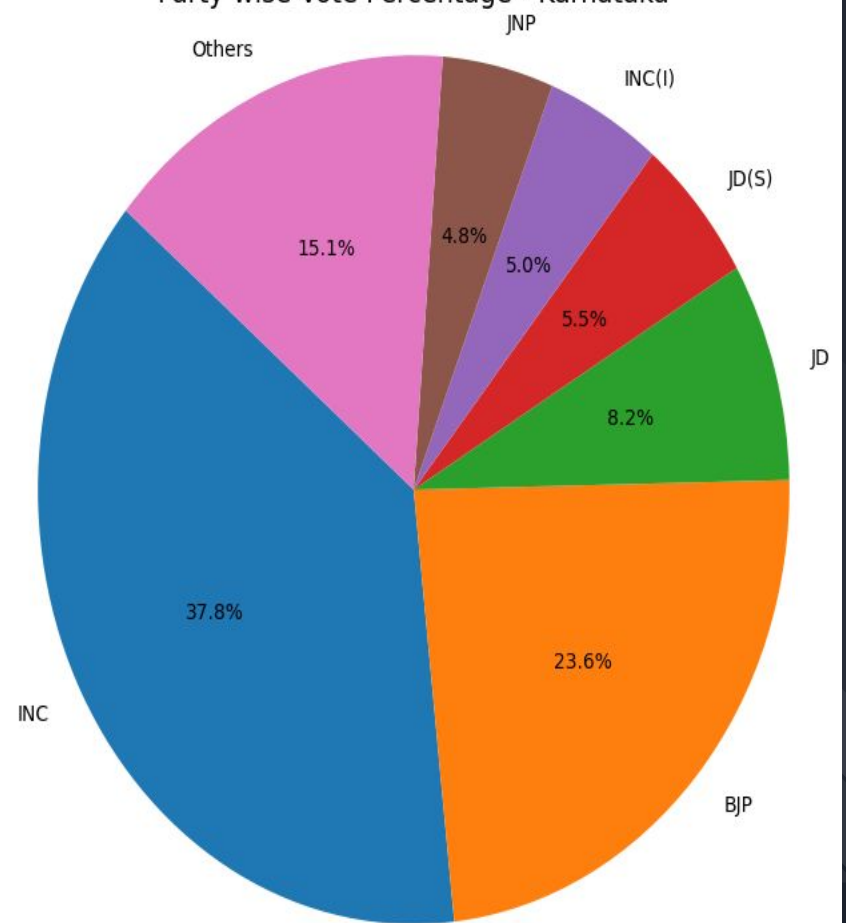
Party-wise Vote Percentage - Gujarat
INC(I) JNP



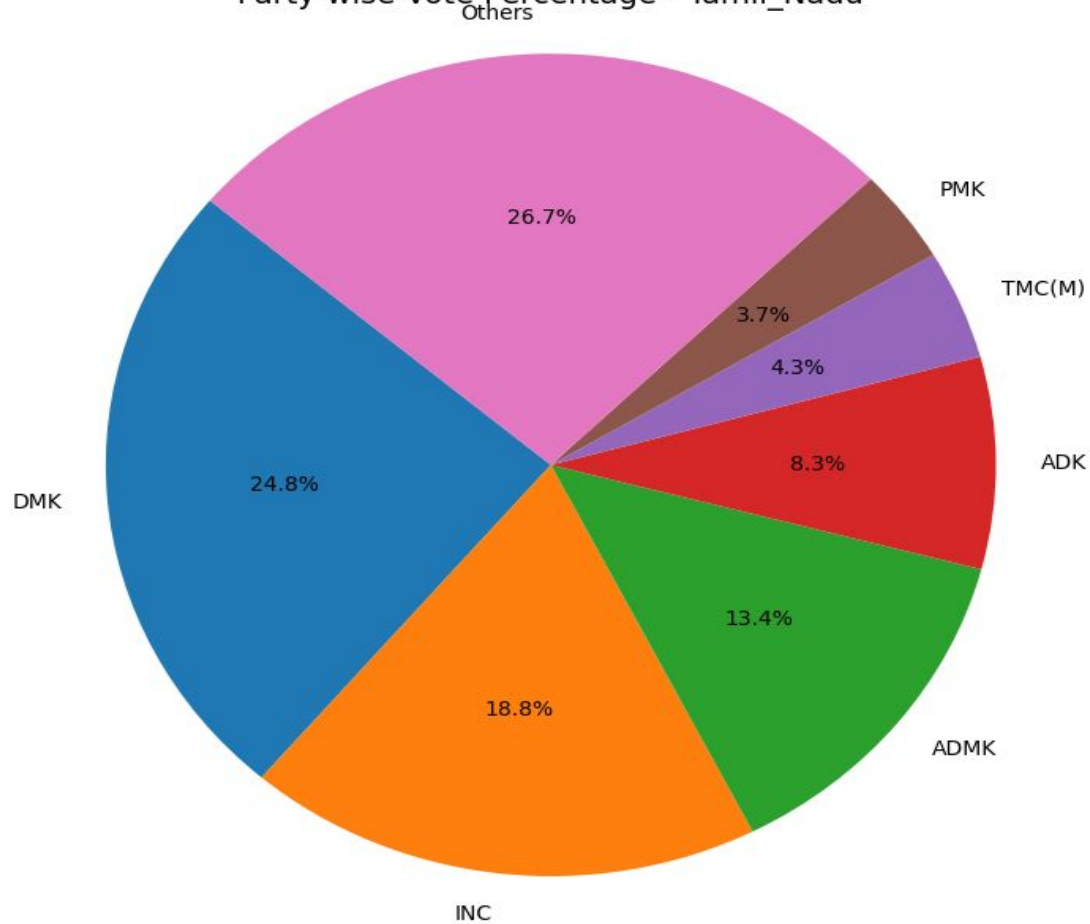
Party-wise Vote Percentage - Rajasthan



Party-wise Vote Percentage - Karnataka



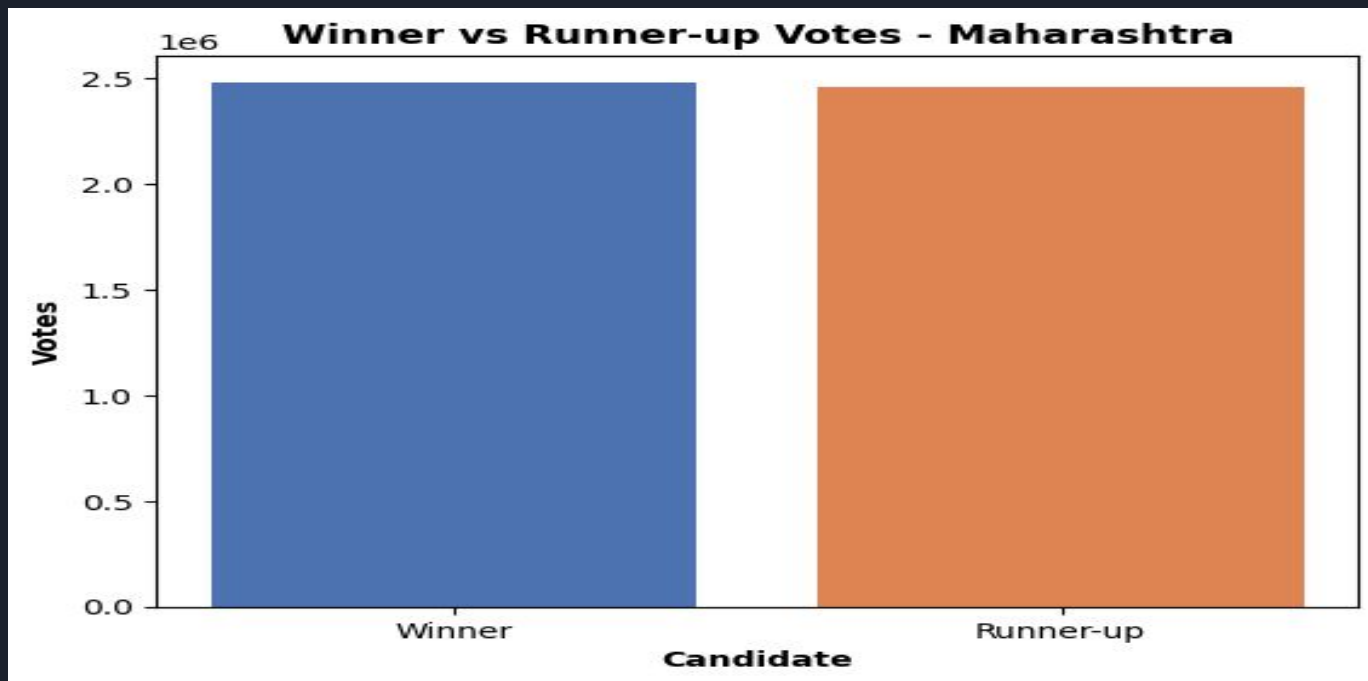
Party-wise Vote Percentage - Tamil_Nadu



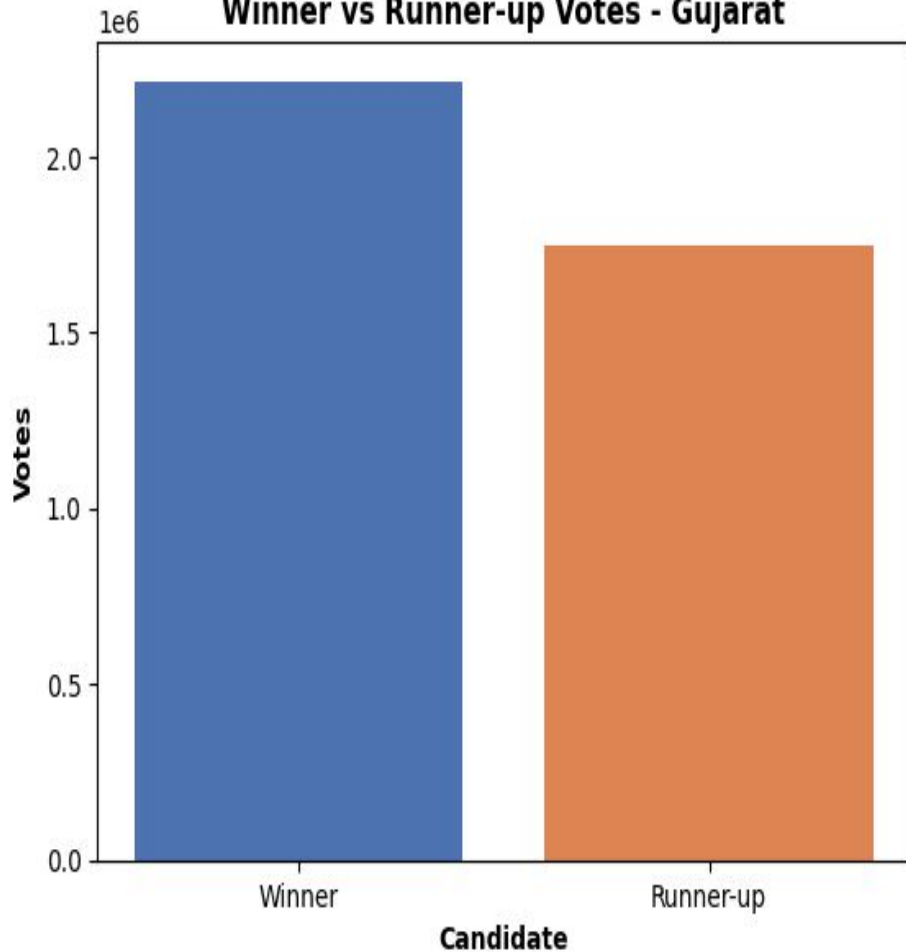
The five pie charts show that voting patterns vary significantly across the selected states. Some states exhibit clear two-party dominance, while others reflect fragmented multi-party competition. Major national parties perform strongly in most states, but regional and smaller parties also play an important role in influencing vote distribution. Overall, the pie charts highlight differences in party dominance, competition level, and voter preferences across states.

2. Winner vs Runner-up Analysis (Bar Graph)

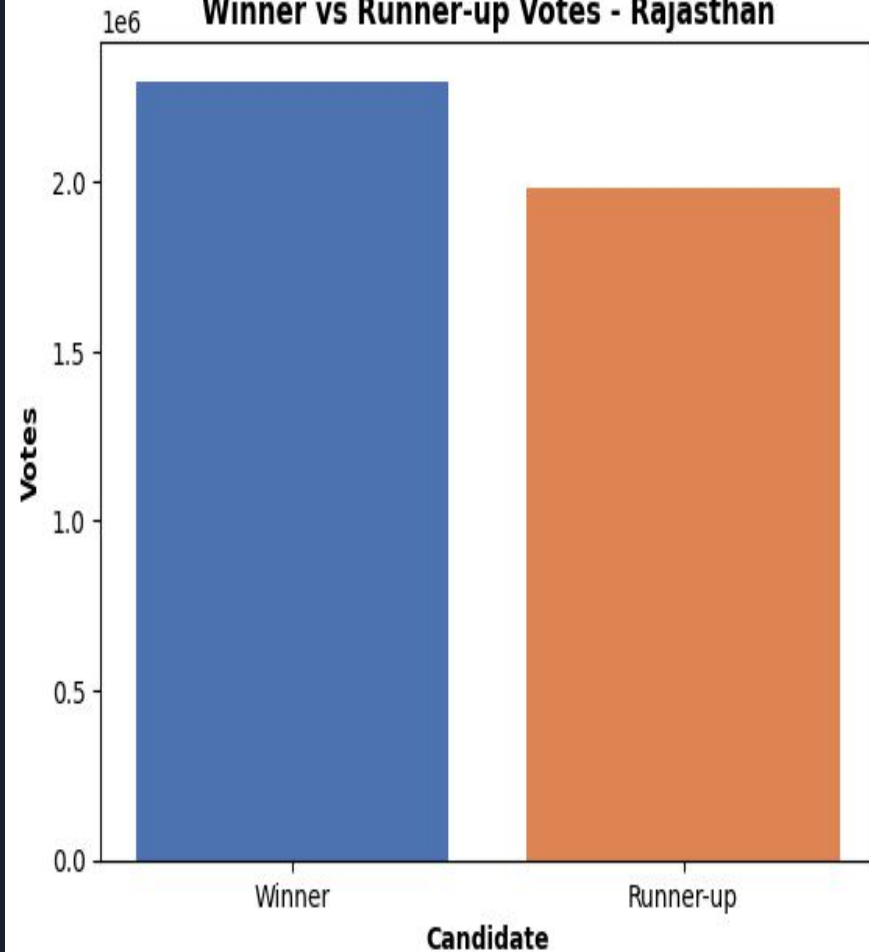
The winner vs. runner-up comparison shows clear differences. Gujarat, Maharashtra, and Tamil Nadu had big winning margins, showing strong mandates, while Karnataka had a close contest, indicating high competition.



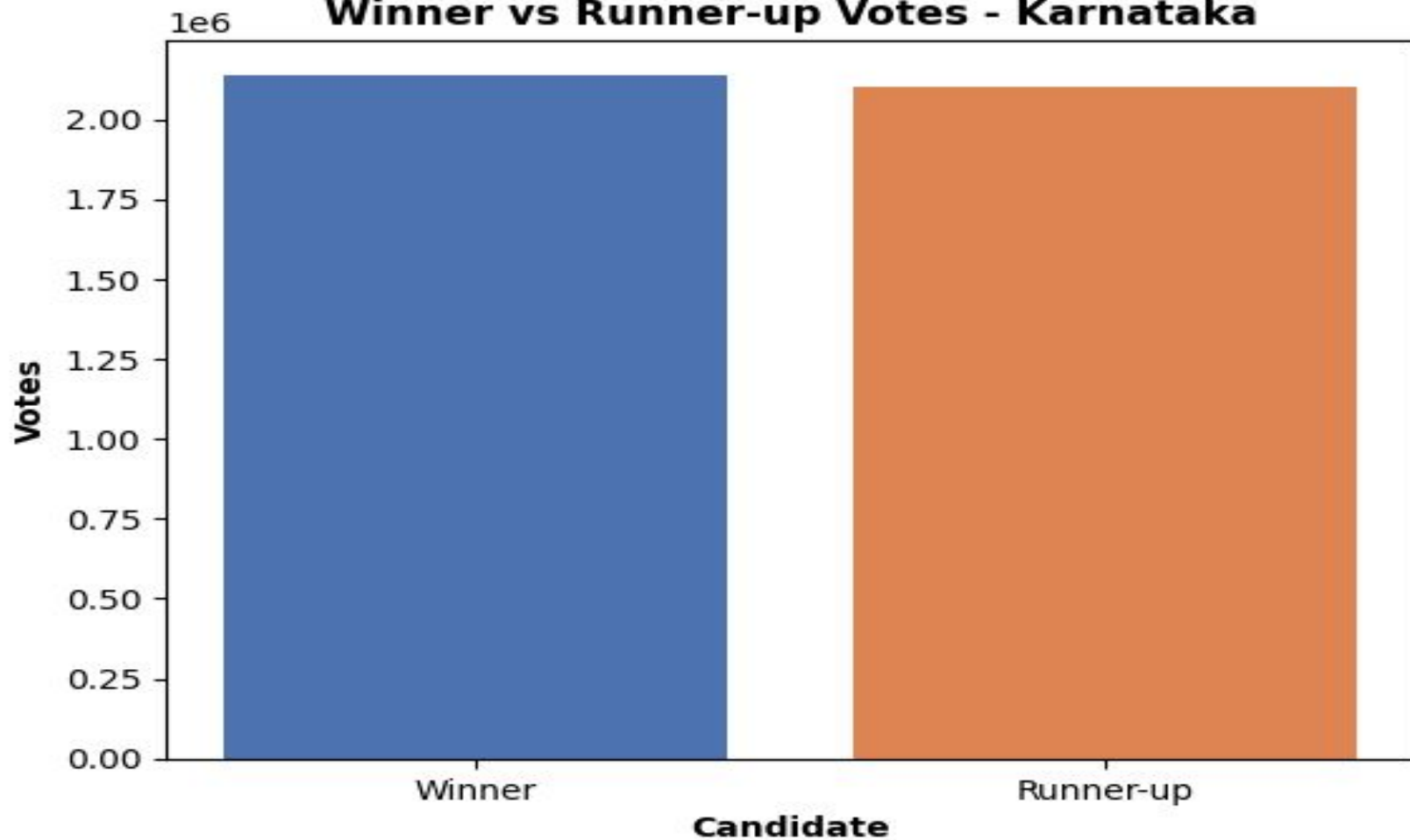
Winner vs Runner-up Votes - Gujarat

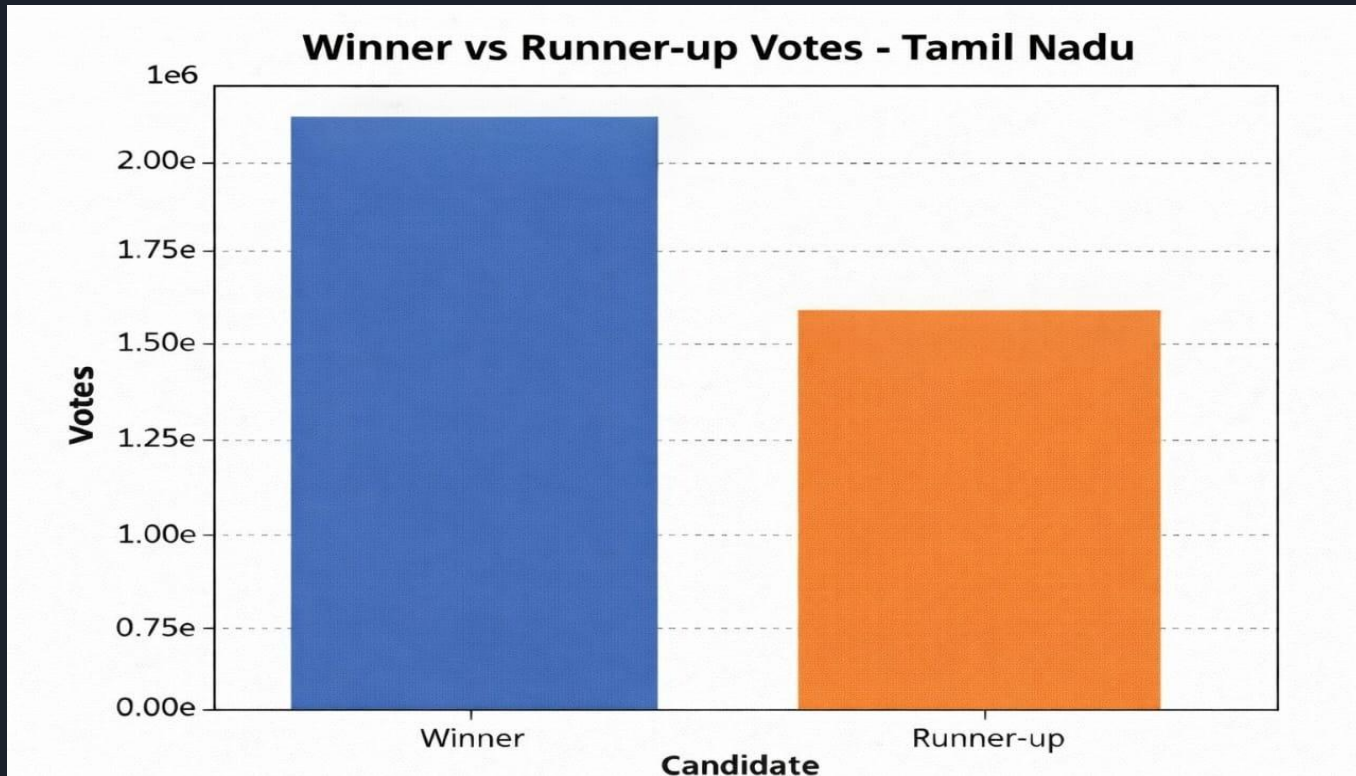


Winner vs Runner-up Votes - Rajasthan



Winner vs Runner-up Votes - Karnataka

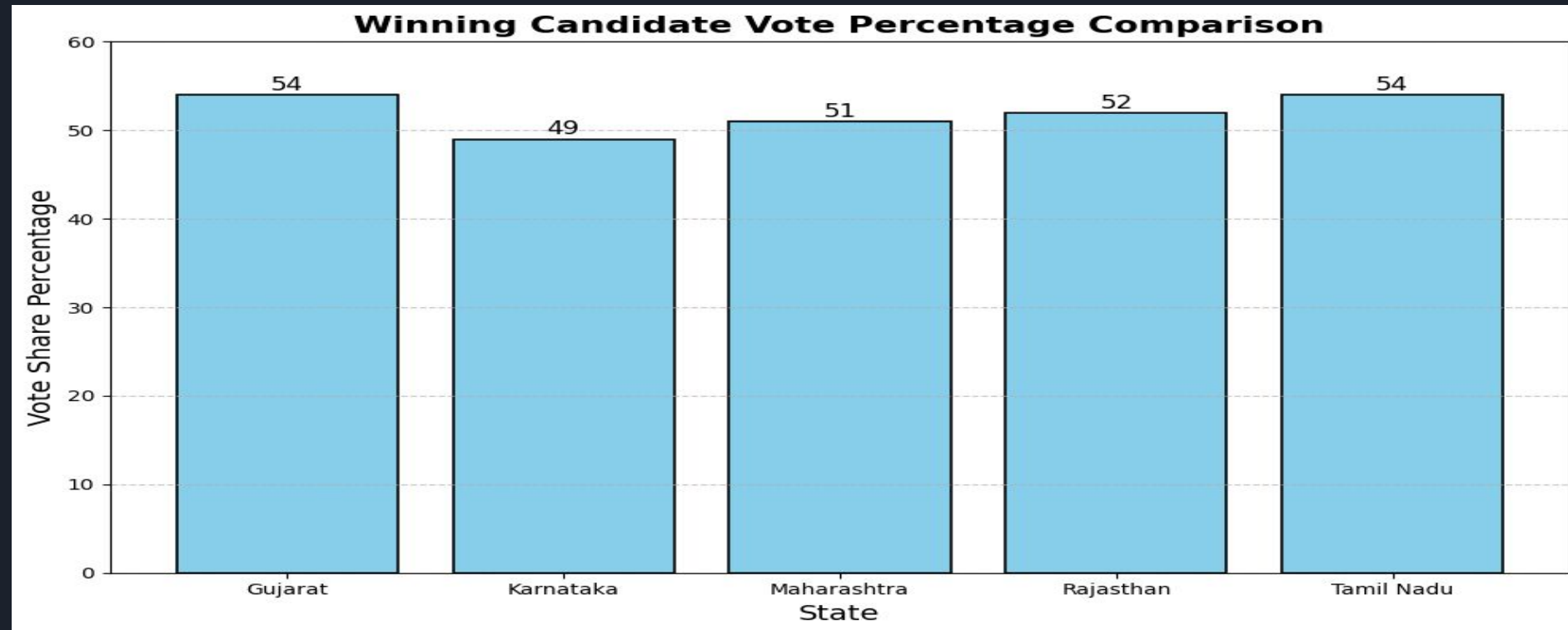




The winner in Tamil Nadu received substantially higher votes than the runner-up, highlighting a decisive victory. This reflects strong public support in the state.

3.State-wise Election Results Overview

This column chart compares winning candidates across different states. It helps identify states with clear victories and those with competitive elections, making it easy to understand overall winning trends and voter support.



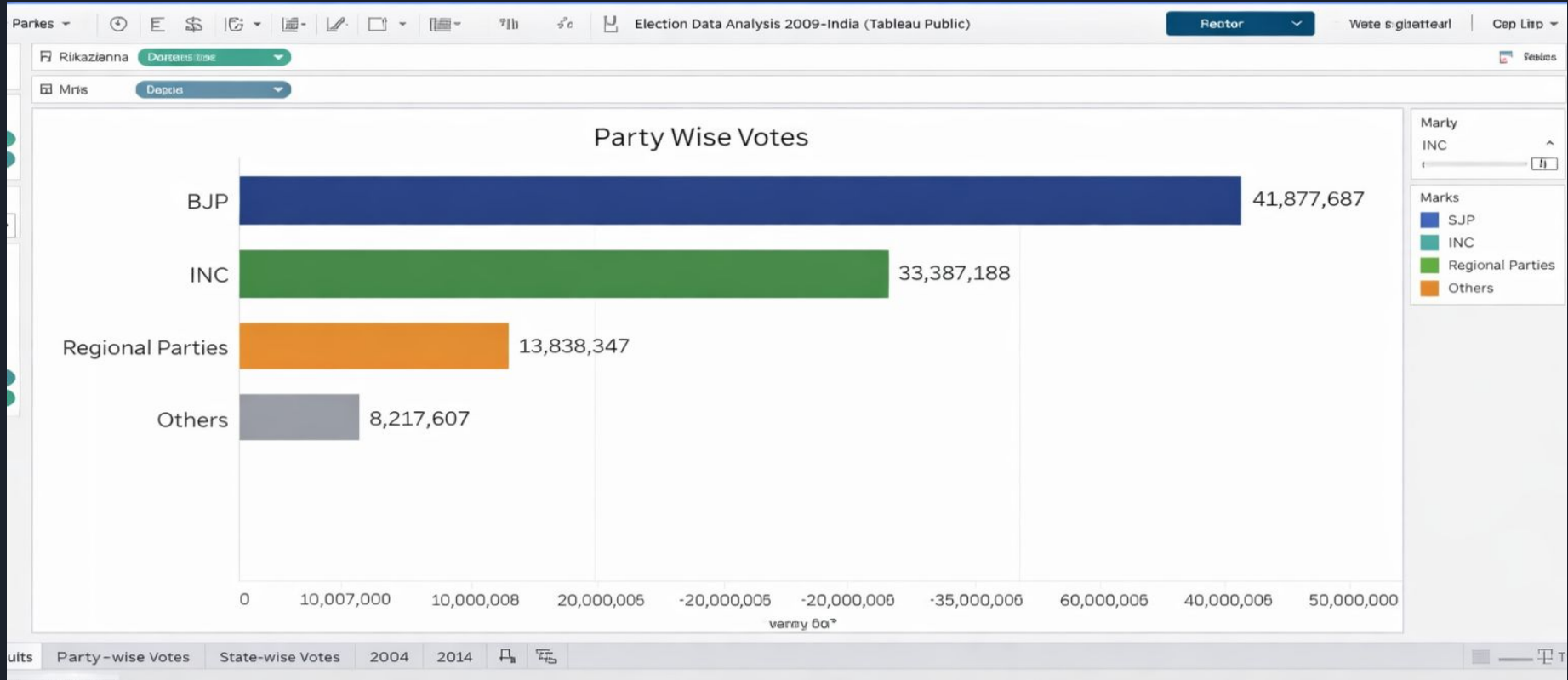
Inferences:-

The chart shows that some states experienced clear and strong victories, while others had closer and more competitive elections. Overall, it highlights differences in voter support across states and helps in understanding the variation in winning trends.

Part 2:- Tableau Visualization

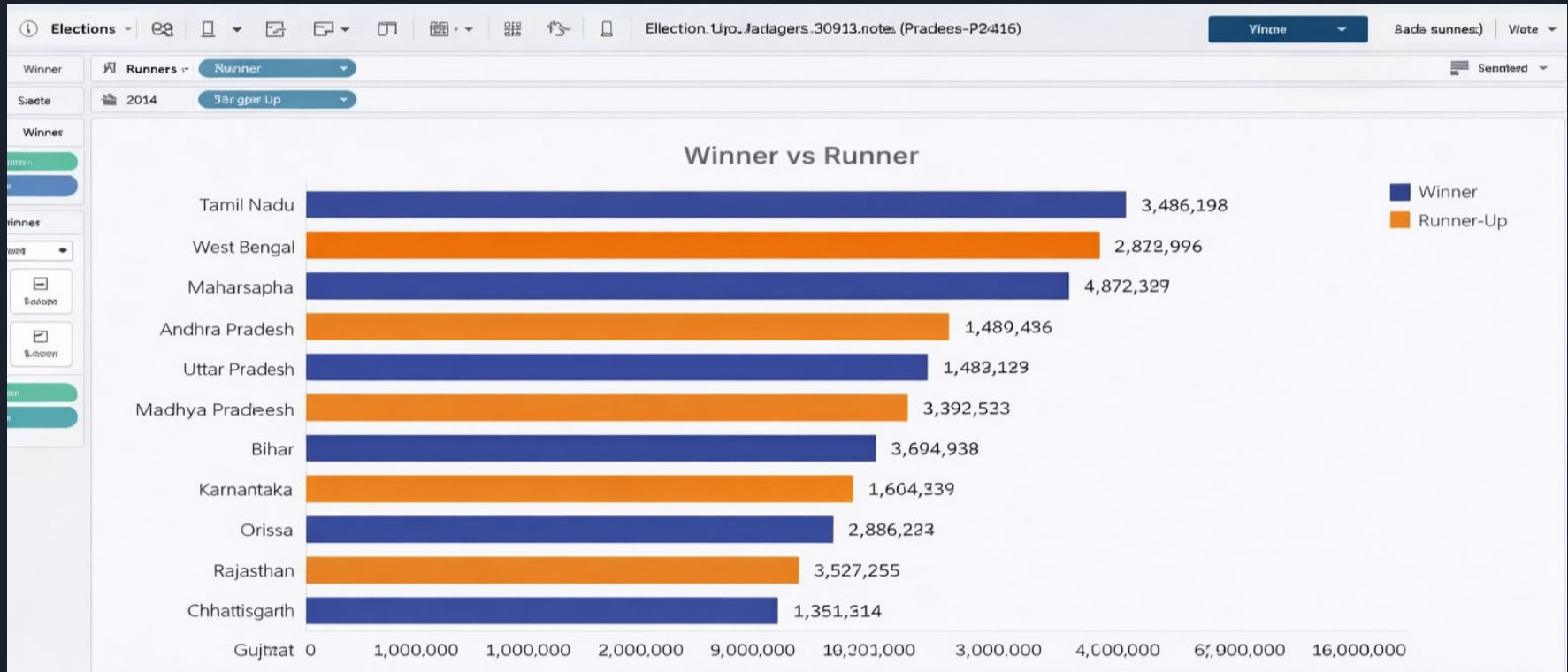
- This Tableau visualization presents election data in a clear and structured way.
- It helps to compare party performance and candidate results easily.
- The charts highlight differences in votes, margins, and percentages.
- Color coding improves understanding and quick comparison.
- Overall, the visualization supports better analysis and decision-making.

Visualization 1:-Party-wise Votes



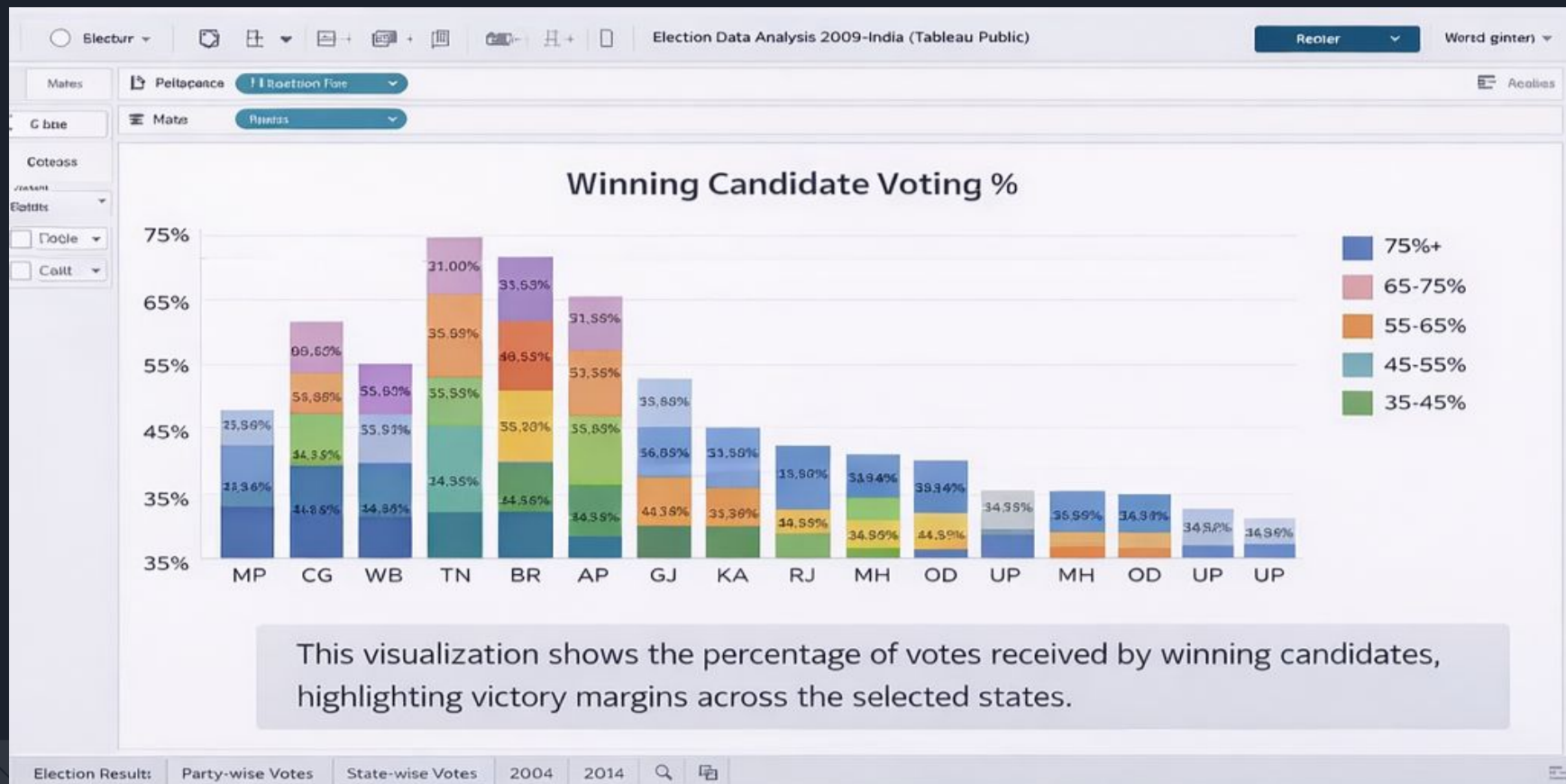
Shows the total votes received by each political party across selected states.
Helps compare overall party performance.

Visualization 2 – Winner vs Runner-Up



This chart Compares votes of the top two candidates in each state.
Highlights the level of electoral competition.

Visualization 3 – Winning Candidates Voting Percentage

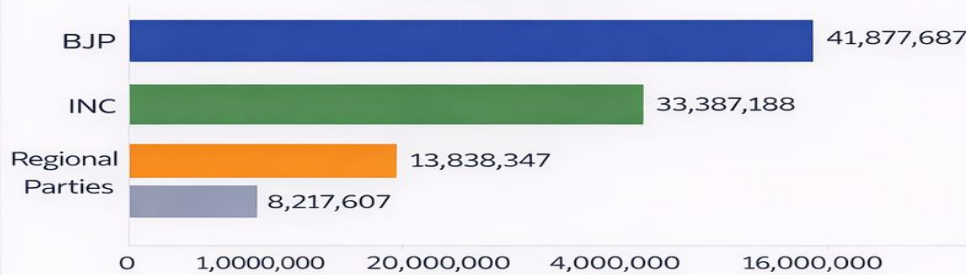


Data visualization

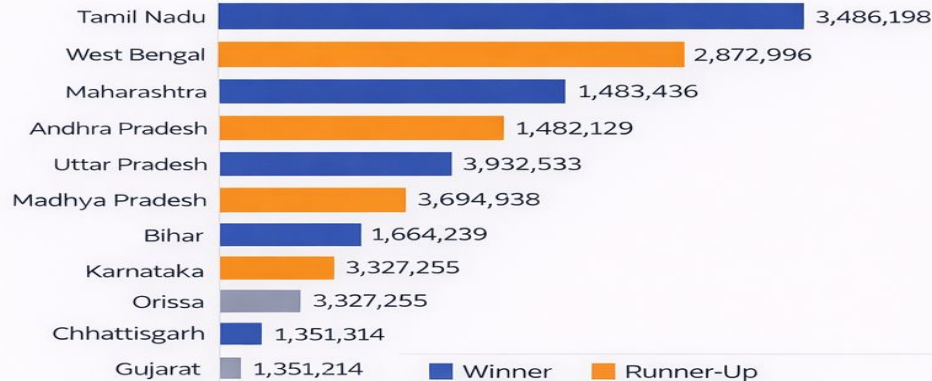
Election Data Analysis Dashboard – India (2009 / 2014)

Year
State

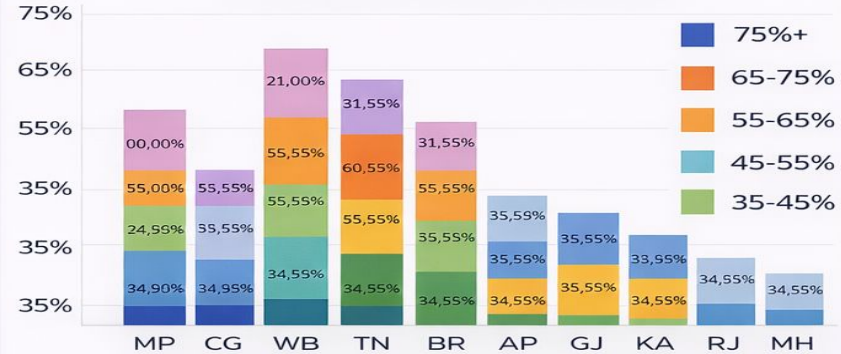
Party Wise Votes



Winner vs Runner-Up



Winning Candidate Voting %



This dashboard provides a comprehensive analysis of Indian election results by combining party-wise vote distribution, winner versus runner-up comparison, and winning vote percentages across selected states.

This three Tableau charts to analyze election results.
It compares party-wise votes, winner vs runner-up margins, and winning vote percentages.
Together, these visuals provide a clear overview of electoral performance and competition.

Part 3:-SQL Analysis (Google BigQuery)

- BigQuery efficiently analyzes large datasets.
- JOIN queries combine multiple tables to extract meaningful insights.
- It handles complex calculations and aggregations easily.
- Trends and patterns can be quickly identified.
- Insights help in faster and more accurate decision-making.



Query

SELECT

a.State,

a.Candidate_Name,

a.Sex,

p.Party_Name,

a.Votes,

a.Margin_Percentage

FROM

`your\_project.your\_dataset.all\_states\_ge` a

JOIN

`your\_project.your\_dataset.ge\_parties` p

ON

a.Party_ID = p.Party_ID

WHERE

a.Year = 2019;

Conclusion:-

This assignment helped us understand election performance using charts, Tableau visualizations, and BigQuery SQL. It showed party-wise distribution, candidate comparisons, and winning vote strengths in Maharashtra, Rajasthan, Karnataka, Gujarat, and Tamil Nadu .Overall, it improved our skills in handling data, creating visualizations, analyzing results, working in a team, and presenting findings.

Thank you